Supplemental Table 1: Search Strategy

NONINVASIVE VENTILATION/ or Continuous Positive Airway Pressure/ or (((noninvasive or (non*1 adj invasive) or hyperbaric) adj3 (ventilat* or respirat* or breath*)) or NBCPAP or nb-cpap or cpap or epap or cppb or cppv or peep or ((expiratory or continuous* or constant) adj2 positive)).tw

DEVELOPING COUNTRIES/ or exp africa/ or exp south america/ or exp central america/ or exp asia central/ or exp asia southeastern/ or exp asia western/ or ((developing or underdevelop*) adj3 (countries or country or nation*1)).tw or ((low*1 or lower* or middle*1) adj2 (country or countries or nation*1)).tw or lmic*1.tw or (third* adj world).tw

(subsahara* or sahara* or angola or benin or Botswana or Burkina or Burundi or Cameroon or cape verde or central Africa or central African republic or chad or comoros or congo or cote d ivoire or democratic republic congo or Djibouti or guinea or Eritrea or Ethiopia or gabon or gambia or Ghana or kenya or Lesotho or Liberia or Madagascar or Malawi or mali or myaotte or Mozambique or Namibia or niger or Nigeria or Rwanda or sahel or Senegal or sierra leone or south Africa or sudan or Swaziland or Tanzania or togo or Uganda or Zambia or Zimbabwe or Somalia or Afghanistan or Albania or Algeria or samoa or argentina or Armenia or Azerbaijan or Bangladesh or Belarus or belize or Bhutan or Bolivia or bosnia or Herzegovina or brazil or Bulgaria or Cambodia or Colombia or costa rica or Croatia or cuba or dominica or Dominican republic or Ecuador or Egypt or Salvador or Fiji or grenada or Guatemala or Guyana or Haiti or Honduras or india or Indonesia or iran or Iraq or Jamaica or Jordan or Kazakhstan or Kiribati or north korea or Kosovo or Kyrgyz or laos or Lebanon or Lesotho or Libya or Macedonia or Malaysia or Maldives or marshall islands or Mauritania or Mauritius or mexico or Micronesia or

moldova or Mongolia or Montenegro or morocco or Myanmar or Burma or Nauru or Nepal or Nicaragua or Pakistan or panama or Paraguay or peru or Philippines or Romania or Russian federation or samoa or Serbia or Solomon or sri lanka or suriname or Syria or Tajikistan or Thailand or timor or togo or tonga or Tunisia or turkey or Turkmenistan or Tuvalu or Uganda or Ukraine or Uzbekistan or Vanuatu or Venezuela or Vietnam or gaza or yemen).tw

Supplemental Table 2: Characteristics of included studies – Non-comparative studies

Author, Year	Country and setting	Study design	Sample size and Population	Intervention and Equipment	Comparison	Outcomes of interest
Balfour-Lynn[16] 2014	Ghana District hospital General ward	Observational implementation study[18]	N = 106 Age 0-5 years, Respiratory distress based on respiratory rate, SpO_2, intercostal retractions, and grunting	NIV via Nippy Junior paediatric pressure controlled portable ventilator	N/A	Mortality* Adverse Events
Bjorkland[17] 2019	Uganda Referral hospital Acute Care Unit	Prospective, non-blinded, non-randomized interventional study	N=83 Age 30 days - 5 years, moderate or severe respiratory distress based on a calculated respiratory score (Tal score > 3) or hypoxia despite low-flow oxygen	SEAL-bCPAP with nasal prong adaptation from ear plug material	N/A	Mortality* Adverse Events* Change in respiratory rate, oxygen saturation, and Tal score[+]
Bonora[18] 2011	Argentina Referral hospital Intensive Care Unit	Retrospective Observational Study	N=154 Age 1-18 years, patients needing NIV for >30 min to attempt to avoid intubation	Neumovent Graph, Neumovent Graph Net, or Harmony devices for NIV	N/A	Mortality Need for intubation* Duration of NIV Duration of hospital stay
Brown[27] 2013	Malawi Referral hospital	Case Report	N=1 Age 6 months, respiratory distress	Low cost bCPAP device developed by authors	N/A	Mortality Adverse Events Vital sign changes after 1 hour Length of hospital stay
Figuera[19] 2017	Argentina Provincial hospital Intermediate Care Unit	Retrospective Descriptive Study	N=120 Age 1-24 months, weight <12 kg, Tal score >5	Hudson RCI-CPAP	N/A	Mortality Adverse Events Success of CPAP (15% decrease in RR) Changes in vital signs and Tal score[+] Duration of NIV Duration of ICU stay
Ghiggi[20] 2000	Argentina Referral hospital	Prospective Observational Study	N=42 Age 1 month- 5 years,	Nasopharyngeal CPAP via Sechrist IV100 B respirators	N/A	Mortality Adverse Events

	Intensive Care Unit		Acute respiratory failure from pulmonary cause with indication for mechanical ventilation			Need for mechanical ventilation* Change in vital signs Duration of NIV
Kinikar[21] 2011	India Referral hospital Intensive Care Unit	Case-control study	N=36 Age 0-12 years, influenza like illness, moderate to severe respiratory distress or respiratory failure	Locally constructed nasal bubble CPAP	N/A	Mortality Adverse Events Changes in vital signs in first 6 hours*
Lum[22] 2011	Malaysia Referral hospital Intensive Care Unit	Prospective Observational Study	N=129 Age 0-16 years, patients deemed likely to require intubation based on vital signs and work of breathing	NIV via Mapleson F breathing system	N/A	Mortality Adverse Events Length of NIV Length of PICU stay Treatment success* (intubation avoided) Vital Sign changes
Machen[23] 2015	Malawi Referral hospital Acute care unit	Prospective Observational Study	N=79 Weight <10 kg, respiratory distress, bCPAP deemed appropriate by physician	bCPAP	N/A	Mortality* Duration of bCPAP Duration of hospital stay Change in RISC score[+]
McCollum[28] 2011	Malawi Referral hospital	Case Report	N=1 3 mon old, respiratory distress	Hudson RCI -bCPAP	N/A	Mortality Adverse Events Duration of bCPAP Change in vital signs
Myers[24] 2019	Malawi Referral hospital Critical care zone, emergency zone	Prospective Observational Study	N=117 Age 0-59 months, severe respiratory distress	Diamedica “Baby CPAP”	N/A	Mortality* Adverse Events
Pulsan[25] 2019	Papua New Guinea Referral hospital	Prospective Observational Study	N=64 Children with severe acute lower respiratory infection, with hypoxaemia or severe	Diamedica-modified Airsep intensity bCPAP	N/A	Mortality Change in respiratory distress score[+]*

	Intensive Care Unit, Special Care Nursery		respiratory distress despite standard oxygen therapy			
Walk[26] 2016	Malawi Referral hospital High dependency unit, emergency ward	Prospective Observational Study	N=77 Age 1 week-14 years, progressive acute respiratory failure despite oxygen and antimicrobial therapy	Locally constructed CPAP	N/A	Mortality* Adverse Events Treatment failure (death or intubation) Duration of CPAP Changes in vital signs

*= primary outcome
[+] = scoring tool to evaluate illness severity
bCPAP=bubble continuous positive airway pressure, CPAP= continuous positive airway pressure, NIV= non-invasive ventilation, RISC score=respiratory index of severity in children, HIV= human immunodeficiency virus, PICU=pediatric intensive care unit, MPSNZ-SS=Modified pediatric society of New Zealand severity score, SA score= Silverman-Anderson score

Supplemental Table 3: Additional findings for Randomized Control Trials

Author, year	Findings
Cam[9] 2002	Stabilization of patient with PaO_2 >80 mmHg after 30 min: CPAP: 14/18 (78%) Oxygen: 6/19 (32%) 13/19 oxygen patients were transitioned to CPAP after failure on oxygen, all improved
Chisti[10] 2015	Treatment failure: bCPAP: 5/79 (6%) Low-flow oxygen: 16/67 (24%) High-flow oxygen: 10/79 (13%) Length of Hospital stay (days; median (IQR)): bCPAP: 5 (3-7) Low-flow oxygen: 4 (3-7) High-flow oxygen: 5 (3-7)
Lal[11] 2018	Decrease in RR at 1 hour (mean, SD): bCPAP: 8 (6) Supplemental oxygen via facemask or hood: 5(4) Need for mechanical ventilation: bCPAP: 2/36 (5%) Standard of Care: 1/36 (3%)
Morales[15] 2004	Duration of Hospital stay (days, mean (SD)): NIV: 8.2 (2.8) Intubation: 19 (11) Success of intervention: NIV: 12 (86%) Intubation: 12 (100%)
McCollum[12] 2019	Duration of respiratory support (days, mean (SD)): bCPAP: 4.5 (1.9) oxygen: 3.9 (2.1)
Wilson[13] 2013	Decrease in RR at 1 hour (mean (CI)): Immediate CPAP: 16 (10, 21) Delayed CPAP: 1 (-2, 5)

	Percent change in RR at 2 hours: Immediate CPAP: data missing Delayed CPAP: 13 (8, 19)
Wilson[14] 2017	Duration of CPAP (median (IQR)): CPAP: 12 (7.2-19.8) Control: 0 (0)

CPAP= continuous positive airway pressure; bCPAP=bubble continuous positive airway pressure; RR=respiratory rate in breaths per minute; AE=adverse events; SD=standard deviation; IQR=interquartile range

Supplemental table 4: Outcomes for non-comparative studies

Author, year	Total Sample Size	Mortality	Additional Findings	Adverse Events	Reported Limitations	Reported Conclusions
Balfour-Lynn[16] 2014	106	2 (2%)	N/A	0 (0%)	Possibility of missing data	NIPPV can be a simple and cost-effective way to treat patients with acute respiratory failure
Bjorklund[17] 2019	83	8 (10%)	Patients with severe illness based on Tal score: 0 hours: 64/83 (77%) 2 hours: 12/83 (15%)	Severe: 0 Mild: 5 (6%) Mild AE included nasal tissue irritation and abdominal distension	Evaluations for complications based only on clinical exam, not powered to evaluate effectiveness, differences in pre-trial and trial patients	SEAL-bCPAP is safe for treatment of respiratory distress in non-neonatal children in LMIC with a trend toward decreased mortality.
Bonora[18] 2011	154	Avoided intubation: 3.8% Required intubation: 38.8%	No need for intubation: 80/154 (52%) Duration of NIV (days, median (IQR)): Avoided intubation: 4 (2.25-6) Required intubation: 2 (1-4) Duration of hospital stay (days, median (IQR)): Avoided intubation: 6 (5-9)	Skin breakdown noted but number of adverse events not reported	Retrospective study design with no control group, no rigid protocol to determine when therapies should be escalated or discontinued	NIV avoided mechanical ventilation in a high proportion of children.

			Required intubation: 13 (9-24)			
Brown[27] 2013	1	0 (0%)	Duration of bCPAP: 4 days Duration of hospital stay: 6 days	0 (0%)	Limitations not reported	A low-cost bCPAP could reduce child mortality in Africa.
Figueroa[19] 2017	120	Not reported	Success of bCPAP: 72% Duration of bCPAP (hrs, mean (CI)): 75 (65-85) Duration of ICU stay (days, mean (CI)): 10 (9-11)	4 (3%) Complications included abdominal bloating and pneumothorax	Limitations not reported	A reduction in respiratory rate, heart rate and TAL scores at 2 hours after starting intervention were predictors of success
Ghiggi[20] 2000	42	2 (5%)	Duration of nasopharyngeal CPAP (days, mean (SD)): 4.12 (3.71) Need for mechanical ventilation: 13/42 (31%)	8 (19%) Complications included tube obstructions and apnea due to excessive sedation	Small sample size	Nasopharyngeal CPAP was useful to avoid mechanical ventilation
Kinikar[21] 2011	36	0 (0%)	Duration of ICU stay (days, median (range)): 2 (2-5) Duration of hospital stay (days,	0 (0%)	Limitations not reported	Indigenous NB-CPAP improves hypoxemia and signs and symptoms in hemodynamically stable children with acute respiratory failure due to influenza-like injury

			median (range)): 7 (6-11) Decrease in mean RR after 6 hours: H1N1 positive: 20 H1N1 negative: 17			
Lum[22] 2011	129	19 (15%)	Duration of NIV (days, median (IQR)): 4 (2-8) Duration of PICU stay (days, median (IQR)): 4.5 (2-9) Avoided mechanical ventilation for ≥5 days: 98 (76%)	29 (22%) Complications included pneumonia while on NIV, pressure from mask, and problems with mask fitting	Not an RCT, no routine use of blood gas sampling, shortage of NIV machines	NIV represents a viable strategy that provided effective respiratory support and prevented intubation in majority of patients.
Machen[23] 2015	79	23 (29%)	Duration of CPAP (days, mean): 3.12 Duration of hospitalization (days, mean): 8.41 Had lower RISC score after 24 hours: 63 (80%)	Not reported	Clinical diagnoses could have led to misclassification	BCPAP was most beneficial to patients with bronchiolitis.
McCollum[28] 2011	1	0 (0%)	Duration of bCPAP (days): 7	0 (0%)	Limitations not reported	BCPAP was successful in treating an infant with PJP pneumonia secondary to HIV infection

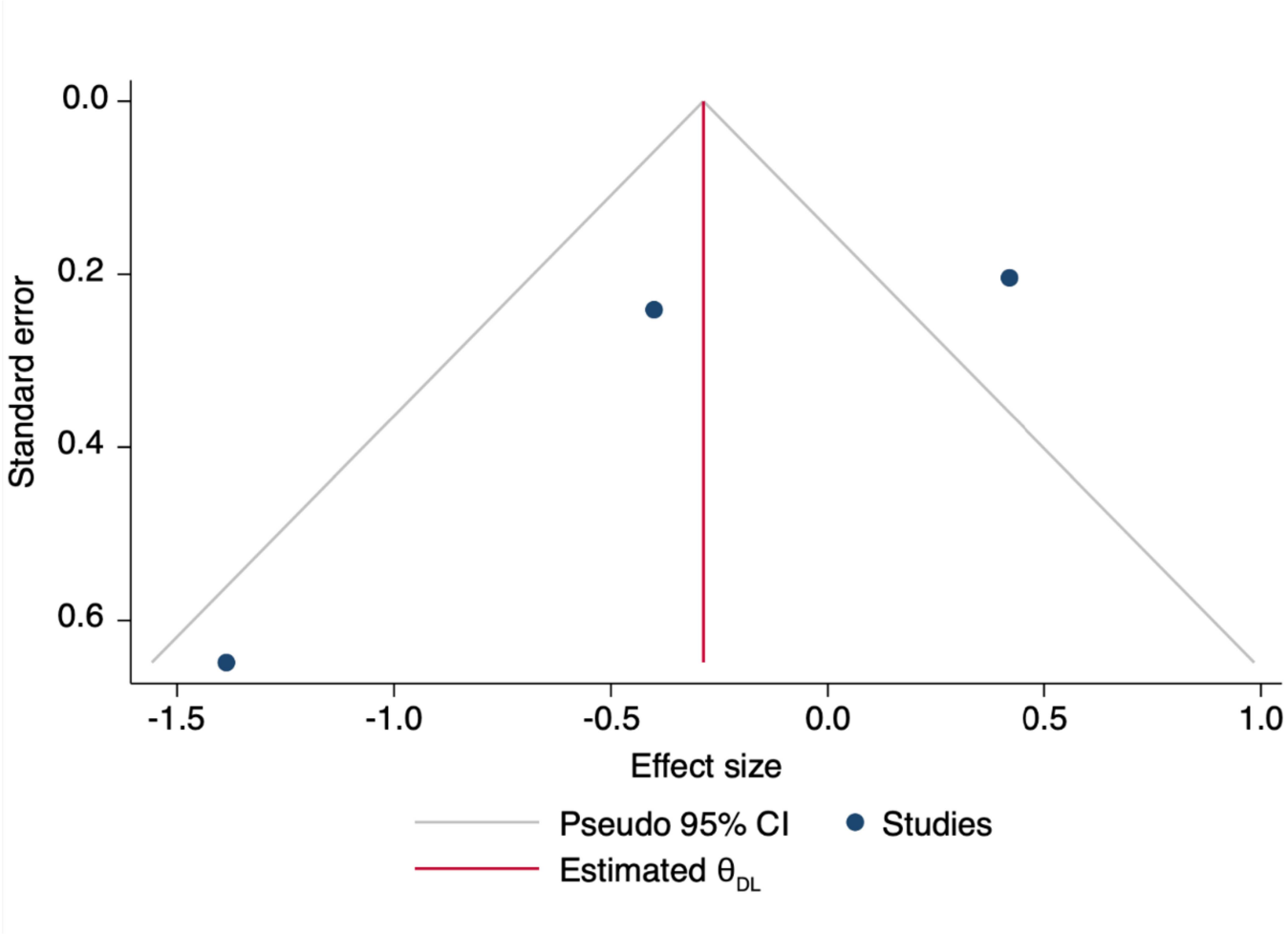
Myers[24] 2019	117	38 (33%)	Required intubation: 15/115 (13%) Duration of treatment (hours, median (IQR)): 24 (24-60)	13 (11%) Complications included blocked nostrils or nasal prongs, interruption of oxygen supply, nasal septum lesions, and aspiration	Observational study design, small sample size, limited human resources and some missing data points,	It is feasible to use bCPAP in the hospital management of critically ill children in resource-limited settings.
Pulsan[25] 2019	64	35 (55%)	RDS (mean (IQR)): Pre-CPAP: 11 (10-12) 1 hour: 9 (8-11) 84 hours: 6.5 (6-8)	Not reported	Observational study design, bCPAP only used when oxygen failed	BCPAP improves oxygenation and reduces respiratory distress in some children but children with comorbidities continue to do poorly
Walk[26] 2016	77	36 (47%)	Duration of treatment (days, median (IQR)): 3 (3-5)	13 (17%)	Non-randomized and uncontrolled, small sample size, understaffing, missing vital sign data	BCPAP can be feasibly implemented into a tertiary African hospital with high-risk patients

CPAP= continuous positive airway pressure; bCPAP=bubble continuous positive airway pressure; RR=respiratory rate in breaths per minute; AE=adverse events; SD=standard deviation; IQR=interquartile range

Supplemental Figure 1. Risk of Bias Assessment for non-comparison studies

Author, Year	Selection	Ascertainment	Causality	Reporting
Balfour-Lynn, 2014				
Bjorklund, 2019				
Bonora, 2011				
Brown, 2013				
Figuera, 2017				
Ghiggi, 2000				
Kinikar, 2011				
Lum, 2018				
Machen, 2015				
McCollum, 2011				
Myers, 2019				
Pulsan, 2019				
Walk, 2016				

Supplemental Figure 2. Funnel plot for meta-analysis of trials assessing CPAP against mortality in children less than five years



Supplemental Figure 3. Funnel plot for meta-analysis of trials assessing CPAP against adverse events in children less than five years

